

Design Document: Modular Memory Management Simulator

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1. Memory Layout and System Assumptions

The simulator provides a high-fidelity abstraction of physical and virtual memory systems. It operates on the following core assumptions:

- **Contiguous Allocation:** Physical memory is modeled as a contiguous byte-array, initialized via the init command.
- **Byte-Level Granularity:** Addressing and allocation are performed at the byte level, utilizing 0-based indexing for physical addresses.
- **Metadata Structure:** Memory is managed using a **Doubly Linked List**. Each node contains a header tracking:
 - Starting Physical Address
 - Block Size
 - Allocation Status (Free vs. Occupied)
 - Unique Process Identifier (PID)

2. Dynamic Partitioning & Allocation Strategies

To satisfy malloc requests, the system implements three standard placement algorithms:

- **First Fit:** Selects the first available hole from the head of the list. Optimizes for allocation speed.
- **Best Fit:** Searches the entire list for the smallest hole that satisfies the request. Optimizes for space efficiency and minimizes "leftover" fragments.
- **Worst Fit:** Selects the largest available hole. Intended to leave a remaining fragment large enough to be useful for subsequent allocations.
- **Block Splitting:** If a selected block exceeds the requested size, the system splits the node into two: one "Occupied" node and a new "Free" node representing the remainder.

3. Deallocation and Memory Coalescing

Upon the free command, the system performs **Immediate Coalescing** to mitigate external fragmentation.

- **Mechanism:** When a block is returned to the "Free" state, the simulator inspects the immediate predecessor and successor nodes in the linked list.
- **Merging:** If neighbors are also free, they are merged into a single contiguous block, effectively recombining fragmented space into larger, more useful segments.

4. Buddy System Design

The simulator includes an optional **Binary Buddy Allocator** for power-of-two memory management.

- **Allocation:** Requests are rounded up to the nearest 2^n size. If a matching block is unavailable, a larger block is split into two "buddies."
- **Recombination:** During deallocation, the system uses XOR address logic to identify a block's unique buddy. If the buddy is free, they are recursively merged.

5. Cache Hierarchy and Replacement Policy

The system simulates a multilevel (L1/L2) set-associative cache hierarchy.

- **Organization:** Parameters include total cache size, line (block) size, and degree of associativity.
- **Replacement Policy (LRU):** The **Least Recently Used** algorithm is used for eviction. The system maintains an access counter/stack; when a set is full, the line with the oldest access timestamp is evicted to make room for new data.

6. Virtual Memory and Address Translation

The virtual memory model utilizes a fixed-size **Paging** system.

- **Translation Pipeline:**
 1. **Extraction:** The Virtual Address is split into a **Virtual Page Number (VPN)** and a **Page Offset**.
 2. **Lookup:** The Page Table maps the VPN to a **Physical Frame Number (PFN)**.
 3. **Synthesis:** $\text{Physical Address} = (\text{PFN} \times \text{PageSize}) + \text{Offset}$.
- **Page Faults:** If the VPN is not mapped (Valid bit = 0), a Page Fault is triggered, simulating the movement of a page from backing store to physical RAM.

7. Limitations and Simplifications

- **Context:** For architectural simplicity, the model assumes a single global page table.
- **I/O Latency:** Disk access during page faults is represented by symbolic latency rather than real-time file I/O operations.
- **Internal Fragmentation:** While external fragmentation is tracked via block headers, internal fragmentation is primarily analyzed within the Buddy System and Paging modules.

Test Artifacts

1. Workload Simulation

The simulator's robustness was validated using three distinct workload types:

1. **Allocation Stress:** Sequential and random malloc calls to evaluate strategy efficiency.
2. **Locality Testing:** Repeated access calls to specific addresses to measure cache hit/miss ratios.

3. **Boundary Analysis:** Testing addresses at the limits of the virtual space to ensure proper error handling.

2. Logs and Trace Analysis

- **Cache Logs:** Tracks hit/miss status and LRU eviction chains.
- **Translation Logs:** Maps Virtual Addresses (VA) to Physical Addresses (PA) (e.g., VA 0x0100 -> Page 0 -> PA 0x0100).

3. Correctness Criteria

1. **Exclusion:** No two "USED" blocks overlap in the memory map.
2. **Merging:** Adjacent free blocks MUST coalesce into a single node.
3. **Persistence:** Recently accessed data must result in a cache hit (Temporal Locality).

Demonstration Scenarios

Feature	Input Command	Expected Observance
First Fit	malloc 1 100 first	System allocates at the first available address (0x0).
Best Fit	malloc 2 50 best	System finds the smallest gap, saving larger holes for future use.
Worst Fit	malloc 3 60 worst	System picks the largest hole to maximize the size of the remainder.
Deallocation	free 1	Marks block ID 1 as free
Coalescing	free 2 access 100	After free2, dump command shows one large FREE block instead of two small ones.
Translation	access 100	Shows address split into Page 0/Offset 100 and mapped to RAM.
Cache LRU	access 10 x2	First access results in a MISS; second results in a HIT.

Physical Memory Allocation & Fragmentation

First Fit, Best Fit, Worst Fit

```
C:\Users\jayja\Documents\memory-simulator>memsim.exe
Memory Simulator Ready. Use 'init memory 1024' to start.
> init memory 1024
Initialized 1024 bytes of physical memory.
> malloc 1 100 first
Allocated at address: 0
> malloc 2 50 best
Allocated at address: 1
> malloc 3 60 worst
Allocated at address: 3
```

```
> free 1
Successfully deallocated ID: 1
> free 2
Successfully deallocated ID: 2
> dump

--- Memory Visualization ---
[0x0000 - 0x0002] FREE Size: 3
[0x0003 - 0x0005] USED (id=3) Size: 3
[0x0006 - 0x03ff] FREE Size: 1018
> █
```

Virtual Memory & Address Translation

```
> access 100
[Page Fault] Mapping virtual page to frame...
L1 Cache Miss. Accessing RAM at: 100
> access 5000
[Page Fault] Mapping virtual page to frame...
```

Cache Hierarchy (LRU Policy)

```
> access 10
[Page Fault] Mapping virtual page to frame...
L1 Cache Miss. Accessing RAM at: 10
> access 10
L1 Cache Hit! Physical Address: 10
> █
```